

FEDERAL BOARD OF INTERMEDIATE & SECONDARY EDUCATION

Annual Examination 2026

Biology: Class 9 (Science Group)

Syllabus: Chapter 1 (The Science of Biology) | Chapter 2 (Biodiversity) | Chapter 3 (The Cell) | Chapter 4 (Cell Cycle) | Chapter 5 (Tissues, Organs and Organ Systems) | Chapter 6 (Molecular Biology) | Chapter 7 (Metabolism, complete)

Total Marks: 38

Time: 1 Hour 45 Minutes

Paper: Exam-17 (Version B)

Instructions:

1. Attempt ALL questions. There is no choice or OR alternative anywhere in this paper.
2. Section A (10 marks): write the question number and the letter of your answer, for example **1. C**. Check that each number matches its letter before you move on.
3. Section B (12 marks): three short questions of 4 marks. Section C (16 marks): two long questions of 8 marks.
4. Answer every part (a), (b), (c)... separately and label it. Where a question asks "why", give the reason, not just the name.
5. Write equations in full with the conditions (for example light and chlorophyll) where the textbook shows them.
6. Use black or blue ink. Pencil is allowed only for the diagram in Question 3(i).

SECTION A: Multiple Choice Questions (10 × 1 = 10 Marks)

#	Q. Choose the best possible option.	A	B	C	D	Answer
1	The starting point of a scientific investigation is:	hypothesis	theory	observation	data	A B C D
2	Information that is gathered as a result of an experiment is called:	theory	observation	hypothesis	data	A B C D
3	If humans and cats belong to the same class, they must also belong to the same:	phylum	order	family	genus	A B C D
4	In a chloroplast, some thylakoids of adjacent grana fuse to form:	cisternae	intergrana	cristae	tonoplasts	A B C D
5	Vitamin D belongs to:	carbohydrates	lipids	proteins	RNA	A B C D
6	According to the textbook, the organ of the human body that uses the maximum energy in the form of ATP (about 25% of the ATP) is the:	brain	heart	skeletal muscles	stomach	A B C D
7	The second phase of photosynthesis is called the light independent reactions because these reactions:	do not depend directly on light	take place only in the dark at night	take place in the thylakoid membranes	release oxygen out of the leaf	A B C D
8	In the electron transport chain, the electrons carried by each FADH ₂ provide enough energy to make:	three ATP molecules	four ATP molecules	two ATP molecules	thirty-six ATP molecules	A B C D
9	Fermentation by yeast is used in:	making yogurt from milk	making cheese	the working of skeletal muscles	wine making and baking	A B C D
10	At the end of the electron transport chain, the final electron acceptor (oxygen) combines with hydrogen ions to produce:	carbon dioxide	water	lactic acid	glucose	A B C D

SECTION B: Short Questions (3 × 4 = 12 Marks)

Q	Question	Marks
2(i)	Write the equation of: (a) photosynthesis (1) (b) alcoholic fermentation (from pyruvic acid) (1) (c) aerobic respiration (1) (d) Name the product of anaerobic respiration in muscle cells. (1)	4
2(ii)	Unripe oranges are green in colour. After ripening their colour changes. (a) Which plastid gives the green colour to unripe oranges? (1) (b) Which plastid gives the orange colour to ripe oranges? (1) (c) Where else in a plant are chromoplasts found, and what is their role there? (1) (d) Give one role of the coloured plastids in ripened fruits. (1)	4
2(iii)	Give four differences between aerobic respiration and anaerobic respiration.	4

SECTION C: Long Questions (2 × 8 = 16 Marks)

Q	Question	Marks
3(i)	Describe and sketch the light dependent reactions of photosynthesis. (a) Draw a labelled sketch of the light dependent reactions. (3) (b) Describe the light dependent reactions step by step. (5)	8
3(ii)	"Meiosis II is identical to mitosis." Explain. (a) Explain the statement by stating what happens to the chromosomes, spindle and nuclear envelope in each phase of meiosis II. (4) (b) Give two ways in which meiosis II is similar to mitosis. (2) (c) Give two ways in which meiosis II is different from mitosis. (2)	8

Invigilator's Signature: _____

Candidate's Signature: _____

End of Question Paper